

Section-A (25 Marks)

1. What is the difference between working stress and limit state design philosophy in the reinforced concrete structures? [5]

Answer:

Working Stress Method and Limit State Method are the two methods used for the design of RCC and steel structures. In the Working Stress Method (WSM), uncertainties in material strength and loads are accounted by using a single overall factor of safety (FOS). The design load is taken equal to actual working loads. It is a deterministic method which assumes that the stresses due to working load don't exceed the safe permissible stresses.

$$\text{Permissible Stress} = \text{Characteristic Strength} / \text{Factor of Safety}$$

Working stress approach does not distinguish between uncertainties arising from loads and those arising from material strengths.

Meanwhile, Limit State Design (LSD) adopts more rational and realistic approach by considering uncertainties in loads and material strengths separately through partial safety factors.

The difference between Limit State Method and Working Stress Method is stated and explained in the points below:

Working Stress Method	Limit State Method
1. Working Stress Method is a traditional and conventional approach of structure design.	1. Limit State Method is a modern approach of structure design.
2. It is a deterministic method which assumes that the stresses due to working load don't exceed the safe permissible stresses.	2. It is a probabilistic approach that considers "Limit States" beyond which the structure ceases to perform.
3. The method is stress-based.	3. The method is strain-based.
4. It utilizes Factor of Safety and produces heavier section.	4. It utilizes Partial Factor of Safety and produces lighter/economical section.
5. Design load is equal to working load in working stress method.	5. Working load is converted into design load by multiplying it with partial FOS.

2. What is the purpose of site investigation? What does a proper site investigation report include? [1.5+3.5=5]

Answer:

Site investigation is the process of acquiring geological, geotechnical and other relevant information which might affect on the design, construction and environment of any civil engineering structure. It is also called site exploration.

Purposes of site investigation:

- Site investigation helps to determine vertical and lateral extension of each soil layer in the site.
- It helps to determine engineering properties of each soil layers through series of in-situ test and lab tests.
- It also helps to determine the position of groundwater table.
- Site investigation also helps to determine index properties of each soil layers through series of lab tests.

Proper site investigation report must consist of:

Chapter 1: Introduction

1.1: Background of project

- 1.2: Objective of investigation
- 1.3: Scope of work
- 1.4: Seismicity of site
- Chapter 2: Methodology
 - 2.1: Boring
 - 2.2: Sampling
 - 2.3: In-situ test
 - 2.4: Observation of GWT
 - 2.5: Lab test
 - 2.6: Analysis theory
 - i. Bearing capacity
 - ii. Scope Stability
 - iii. Liquefaction potential
- Chapter 3: Findings
 - 3.1: Strata
 - 3.2: GWT
 - 3.3: Engineering properties
 - 3.4: Classification of soil
- Chapter 4: Calculation and analysis
 - i. Bearing capacity
 - ii. Scope Stability
 - iii. Liquefaction potential
- Chapter 5: Recommendation
 - i. Size, shape and depth of foundation with bearing capacity
 - ii. Preventive measures for slope stability liquefaction

3. Discuss the structural design consideration for bridges in hydropower projects particularly in hilly and riverine environment. [10]

Answer:

Bridges in hydropower projects provide access to dams, powerhouses, tunnels, and other project facilities. Their design must ensure strength, stability, durability, and safety under difficult terrain and hydraulic conditions.

The structural design consideration for bridges in hydropower projects particularly in hilly and riverine environments are given below:

- i. Site investigation:
 - Detailed topographical, geological, geotechnical, and hydrological surveys are conducted.
 - Foundation conditions, rock quality, groundwater, and slope stability are assessed.
 - The river characteristics, flood levels, and sediment transport are also determined.
- ii. Hydraulic considerations:
 - Design flood is determined.
 - Adequate waterway must be provided to pass floods without excessive afflux.
 - The scour depth is determined and foundations must be provided below maximum scour level.
 - The debris, driftwood, boulders, and floating logs during floods must be considered.
- iii. Foundation Design:
 - The suitable foundation type (open, well, pile, or rock anchors) must be selected based on soil and rock conditions.
 - The resistance must be ensured against scour, erosion, uplift, sliding, and settlement .
 - In hilly regions, foundations are often placed on sound rock.
- iv. Structural loading:

➤ The bridge is designed for:

- Dead load
- Live load (vehicular/pedestrian)
- Impact load
- Wind load
- Seismic load (Nepal is in a high seismic zone)
- Temperature effects
- Braking and traction forces
- Hydraulic forces and debris impact

v. Seismic Design:

- Ductile detailing is done in accordance with seismic design codes.
- Seismic bearings and expansion joints are provided where necessary.
- Adequate lateral stability and redundancy must be ensured.

vi. Bridge type selection:

- The bridge type is chosen based on span and site conditions:
 - Beam/Slab bridges for short spans.
 - Steel truss bridges for medium to long spans.
 - Arch bridges in narrow rocky gorges.
 - Suspension or cable-stayed bridges where long spans are required and river foundations are difficult.

vii. Durability:

- High-quality reinforced or prestressed concrete and corrosion-resistant steel are used.
- Proper concrete cover and waterproofing must be ensured.
- Protection must be provided against abrasion from sediments and corrosion due to moisture.

viii. Hilly terrain considerations:

- Slope stabilization is done using retaining walls, rock bolts, shotcrete, and drainage systems.
- Protection works must be constructed against landslides, rockfalls, and erosion.
- The excavation must be minimized and unstable slopes must be avoided.

4. Explain the importance of shear strength in foundation design and slope stability. Discuss how water content affects the shear strength of soil? [5+5=10]

Answer:

The maximum resistance offered by soil against shearing or sliding along internal planes is known as shear strength of the soil. It is one of the most important engineering properties of soil as it determines the stability and safety of foundations, slopes, retaining structures, and earth dams. The shear strength of soil expressed by Mohr-Coulomb equation:

$$\tau = c + \sigma' \tan \phi$$

where :

τ = Shear strength

c = Cohesion of soil

σ' = Effective normal stress

$$\phi = \text{Angle of internal friction}$$

Importance of shear strength in design of foundation:

- i. **Determination of Bearing Capacity**
 - a. Shear strength governs the ultimate bearing capacity of the soil.
 - b. Higher shear strength allows the soil to support heavier loads safely.
- ii. **Prevention of Shear Failure**
 - a. Foundations are designed such that the applied load does not exceed the soil's shear strength.
 - b. It prevents general, local, or punching shear failure.
- iii. **Selection of Foundation Type**
 - a. Soils with high shear strength can support shallow foundations.
 - b. Weak soils with low shear strength may require deep foundations (piles or caissons).
- iv. **Settlement Control**
 - a. Adequate shear strength minimizes excessive settlement and differential settlement.
- v. **Safe and Economical Design**
 - a. Accurate estimation of shear strength leads to economical foundation dimensions while ensuring safety.

Importance of shear strength in slope stability:

- i. **Resistance to Sliding**
 - a. Shear strength provides resistance against downslope movement of soil masses.
- ii. **Factor of Safety**
 - a. Slope stability depends on the balance between driving forces and resisting shear strength.
 - b. A higher shear strength results in a greater factor of safety.
- iii. **Design of Safe Slopes**
 - a. It helps determine the safe slope angle and height for embankments, cuttings, and earth dams.
- iv. **Prevention of Landslides**
 - a. In hilly regions like Nepal, knowledge of shear strength is essential for designing stable road cut slopes and preventing landslides.
- v. **Effect of Groundwater**
 - a. Increased pore water pressure reduces effective stress and consequently decreases shear strength, making slopes more susceptible to failure.

Section-B (25 Marks)

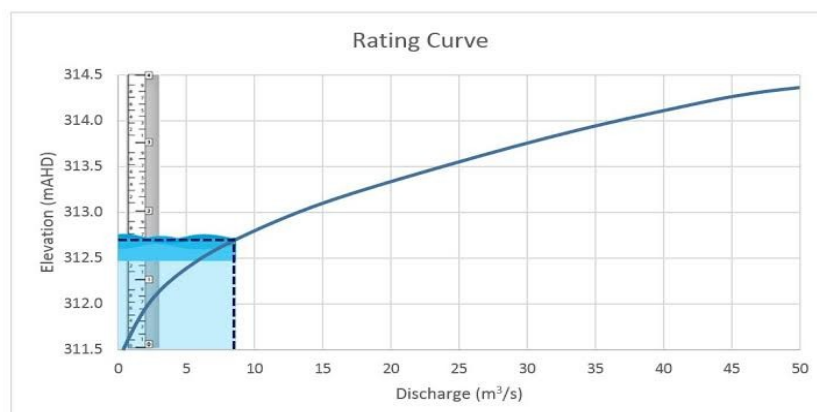
5. Describe the procedure for developing a rating curve. [5]

Answer:

The graph representing the relation between stage and discharge is called rating curve.

The measured value of discharges when plotted against the corresponding stages gives the relationship that represents the integrated effect of a wide range of channel and flow parameters. The combined effect of this parameter is termed control.

If G- Θ relationship for a gauging station is constant, the control is said to be permanent. If it changes with time, it is called shifting control.



The procedure for developing a rating curve is described below:

- i. **Select a Suitable Gauging Site:**
 - a. Straight, stable river reach with a uniform cross-section is selected.
 - b. Bends, backwater effects, and excessive sediment deposition or erosion must be avoided.
- ii. **Install a Water Level Gauge**
 - a. A staff gauge or automatic water level recorder is installed to measure river stage accurately.
 - b. A permanent reference benchmark should be established.
- iii. **Measure Discharge at Different Stages**
 - i. Discharge measurements are conducted over a wide range of flow conditions (low, medium, and high flows).
 - b. **Velocity-area method** with a current meter or an Acoustic Doppler Current Profiler (ADCP) is used.
- iv. **Record Corresponding Stage and Discharge**
 - a. For each discharge measurement, the corresponding water level (stage) is recorded.
 - b. Sufficient data is collected throughout different seasons.
- v. **Plot Stage vs. Discharge**
 - a. **Stage (H)** on the x-axis and **discharge (Q)** on the y-axis is plotted.
 - b. A smooth best-fit curve is drawn through the plotted points. This curve is called the rating curve.
- vi. **Develop the Rating Equation**
 - a. Fit an empirical equation to the curve, generally:
$$Q = C(H - a)^n$$
where:
Q = Discharge (m³/s)
H = Gauge height (stage)
a = Gauge reading corresponding to zero flow
C, n = Empirical constants
- vii. **Verification and Updating**
 - a. Periodically the curve is verified by taking new discharge measurements.
 - b. The rating curve is revised if changes occur due to scour, sediment deposition, vegetation growth, or changes in the river channel.

6. Discuss the type of hydropower projects in Nepal. Explain the layout and major components of a typical run-of-river hydropower plant. Describe the concept of load curve and capacity factor in power systems. [4+6=10]

Answer:

Hydropower projects in Nepal are classified into following types.

On the basis of storage capacity:

- i. **Run-of-river plants:**
 - RoR hydropower plants do not regulate the hydrograph of the river by one or more seasons.
 - The plant utilizes the minimum flow in a river having no appreciable pondage at its upstream.
 - In this type of plants, weir or barrage construct across a river, simply to rise and maintain the water level at certain height.
 - Usually, low head (up to 70 m) scheme and is suitable only on a perennial river with sufficient dry weather flow to generate designed power.
 - Limited storage is used to meet hourly or weekly function of flow.
 - Eg: Khimti HPP, Bhotekoshi HPP etc.
- ii. **Peaking run-of- river plants:**
 - RoR plants constructed with pondage which can regulate daily or weekly hydrograph to run the plant under full capacity. This type of plant is known as peak run-of-river plant.
 - They are constructed to withstand increased load during peak hours.
 - Eg: Marsyangdi HPP, Middle Marsyangdi HPP, Kaligandaki A HPP, Sunkoshi HPP etc.

iii. Storage plants:

- Those plants which regulate the hydrograph of the river by one or more seasons are termed as storage plants.
- Powerhouse is located at the foot of dam or away from the dam.
- Water is carried through tunnel to powerhouse.
- It is suitable for medium to high head projects.
- Eg. Kulekhani HPP

iv. Pump storage plants:

- Plants having water reservoir at upstream as well as downstream of powerhouse, which collects water in upstream reservoir by pumping water from downstream reservoir using cheaper energy.
- Energy is generated during peak hours.

On the basis of head:

i. Low head plant:

Head is less than 15 m.

Axially flow turbine is used in low head plant. E.g., Kaplan turbine

ii. Medium head plant:

Head is between 15m to 50 m.

Mixed flow turbine is used. E.g., Francis turbine

iii. High head plant:

Head is greater than 50 m.

High turbine is used. E.g., Pelton turbine

On the basis of capacity:

i. Micro hydropower project:

Capacity is up to 100 kW.

ii. Mini hydropower project:

Capacity is between 100 kW to 1000 kW.

iii. Small hydropower project:

Capacity is between 1 MW to 25 MW.

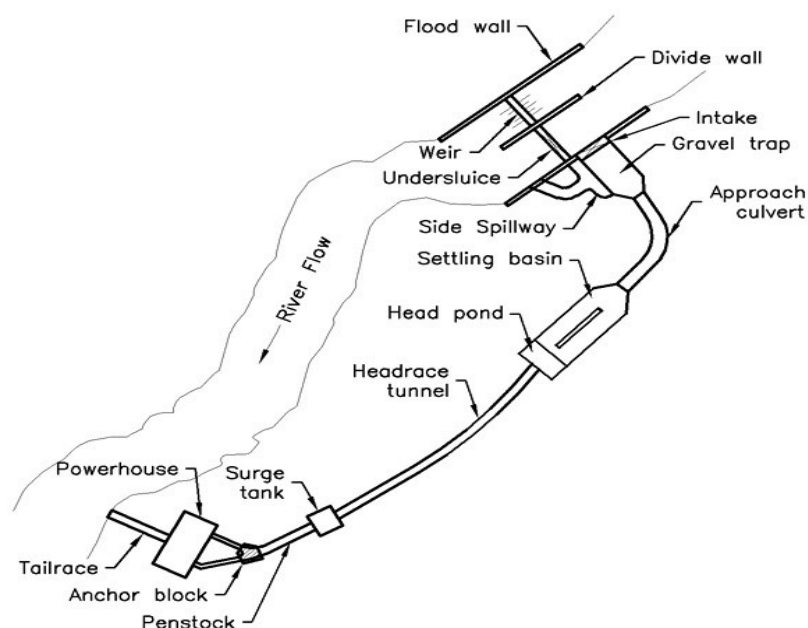
iv. Medium hydropower project:

Capacity is between 25 MW to 100 MW.

v. Large hydropower project:

Capacity is greater than 100 MW.

The layout and major components of a typical run-of-river hydropower plant are given below:



Components:

i. Weir:

- It is constructed across the river to divert the part of river flow to various water conveyance systems through the intake.

ii. Intake:

- It is a structure constricted to divert the required amount of water into a conduit leading to the power plant.
- Trash racks are provided to prevent entry of floating debris and coarse bed load into water conveyance systems.

iii. Gravel trap:

- It is required to flush out bed sediments that enter the approach canal back into a river.
- The gravel trap is constructed close to the intake in order to prevent gravel from getting into the approach channel.

iv. Settling basin:

- It is the structure to remove suspended sediments from the conveyance water for power plant.
- It shall be designed to ensure that the water entering the water conveyance system is free from sediments that can damage the penstock and turbine runners.

v. Headrace conveyance:

- A canal or pipe or tunnel or combination of these are provided for delivering the design flow to the forebay or surge tank with a minimum energy loss.

vi. Forebay:

- It is located at the beginning of penstock shaft that supplies required flow to the turbine during start up, accommodate the reject flow during shut down, reduce water hammer effect.
- It serves as a secondary settling basin as well as a sediment sluice to remove settle suspended particles.

vii. Surge tank:

- It is located between the headrace pressure conduits and the steeply sloping penstock pipe.
- It is generally a cylindrical storage reservoir connected to penstock pipe close to turbine.

viii. Penstock:

- It is a pipe that conveys the flow from forebay or surge tank to the turbine.
- They are designed to carry water to turbines with the least possible loss of head consistent with the overall economy of the project.

ix. Anchor blocks:

- It is an encasement of a penstock designed to restrain the pipe movement in all durations.
- Anchor block should be placed at all sharp horizontal and vertical bends, as there are forces at such bends which tends to move the pipe out of the designated alignment.

x. Powerhouse:

- It is a structure complex where all the equipment for providing electricity is suitably arranged.
- Two requirements of powerhouse planning are functional efficiency and aesthetic beauty.

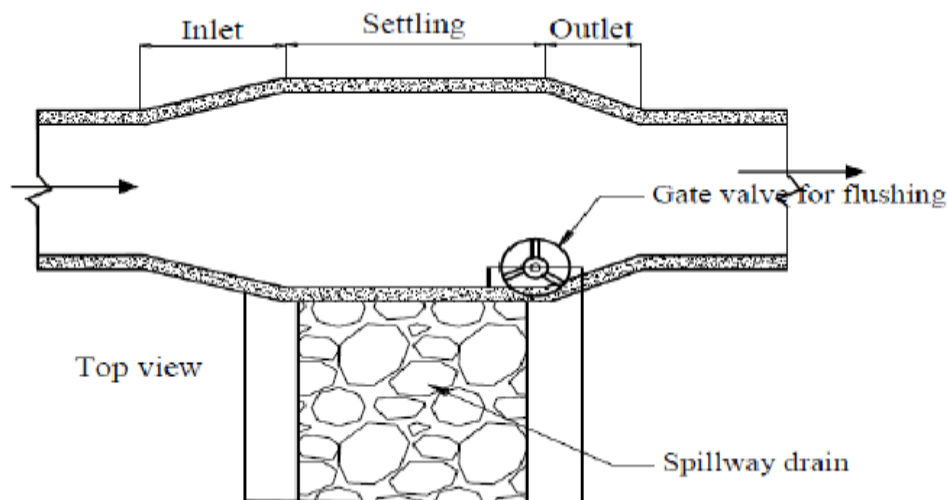
xi. Tailrace:

- It disposes the design flow discharged by hydraulic turbine after the power generation.

7. Discuss the key design considerations of de-sanding basin for a run-of-river project in Nepal. How do suspended sediment characteristics influence basin dimensions and flushing efficiency? What operational challenges arise during monsoon? [4+3+3=10]

Answer:

De-sanding basin also known as de-stilling basin is a hydraulic structure placed between intake and penstock in the hydropower project for removal of suspended sediment particles from flowing water which cannot be trapped by the gravel trap.



The design considerations for sediment flushing in the de-sanding basin are listed below:

- i. **Flushing velocity:**
 - The flushing velocity must be high enough for the transportation of settled sediment out of settling basin.
 - Usually, flushing velocity of 2-3 m/s must be maintained during flushing.
- ii. **Flushing channel:**
 - Flushing channels must be provided at the bottom of settling basin for the efficient collection and removal of sediments.
- iii. **Bottom slopes:**
 - Adequate longitudinal and transverse slope are provided to facilitate efficient movement of sediments towards flushing outlet.
- iv. **Flushing gates:**
 - Flushing gates should be located at the suitable positions and designed to pass sediment-laden water quickly and efficiently.
- v. **Water requirement:**
 - Sufficient water discharge is required to generate required flushing velocity.

Suspended sediments carried by rivers significantly affect the design and operation of de-sanding basins, in hydropower and irrigation projects.

Suspended sediment characteristics influence basin dimensions as:

- i. **Particle Size**
 - a. Coarse particles settle quickly and require a smaller basin.
 - b. Fine particles (silt and clay) settle slowly and require a longer basin with greater detention time.
- ii. **Sediment Concentration**
 - a. High sediment concentration requires a larger basin to provide adequate storage and settling capacity.

- b. Greater sediment loads increase the frequency of flushing.
- iii. **Specific Gravity and Density**
 - a. Heavier particles settle faster than lighter particles.
 - b. Basins handling lighter sediments require larger dimensions.
- iv. **Settling Velocity**
 - a. Basin length and depth are determined based on the settling velocity of the design particle.
 - b. Lower settling velocity requires increased detention time and, therefore, larger basin dimensions.
- v. **Flow Velocity**
 - a. Flow velocity inside the basin should be low enough to allow particles to settle but high enough to prevent deposited sediments from being re-suspended.

Suspended sediment characteristics influence basin dimensions as:

- Coarse sediments are flushed easily through flushing channels.
- Fine sediments tend to remain suspended or adhere to the basin floor, reducing flushing efficiency.
- Efficient flushing requires:
 - Adequate flushing velocity.
 - Properly designed flushing channels and gates.
 - Smooth basin floor with suitable longitudinal slope.

Operational challenge during monsoon season:

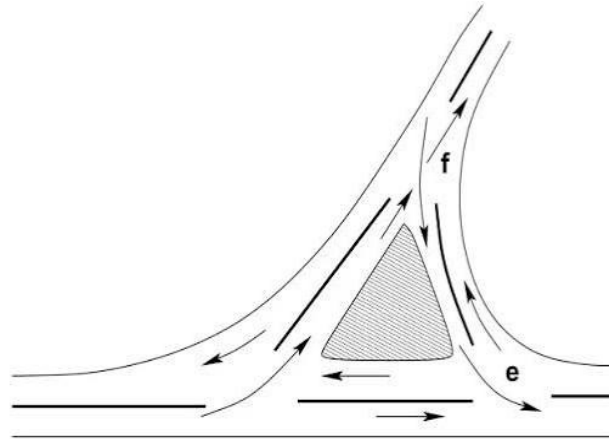
- i. **Very High Sediment Load**
 - a. Rivers carry large quantities of sand, silt, gravel, and debris during monsoon.
 - b. Basins fill rapidly, reducing sediment removal efficiency.
- ii. **Frequent Flushing Requirement**
 - a. Increased sediment deposition necessitates frequent flushing, leading to operational interruptions.
- iii. **Reduced Hydraulic Efficiency**
 - a. Excessive deposition reduces the effective basin volume and detention time.
- iv. **Abrasion of Structures**
 - a. High sediment concentration causes severe abrasion of gates, turbines, spillways, and other hydraulic structures.
- v. **Clogging by Debris**
 - a. Driftwood, vegetation, and floating debris may block inlet structures and flushing outlets.
- vi. **High Flow Velocity**
 - a. Flood flows increase turbulence, reducing sediment settling efficiency and causing re-suspension of deposited particles.
- vii. **Maintenance Difficulties**
 - a. Continuous sediment removal and inspection are difficult during periods of high river discharge.

Section-C (25 Marks)

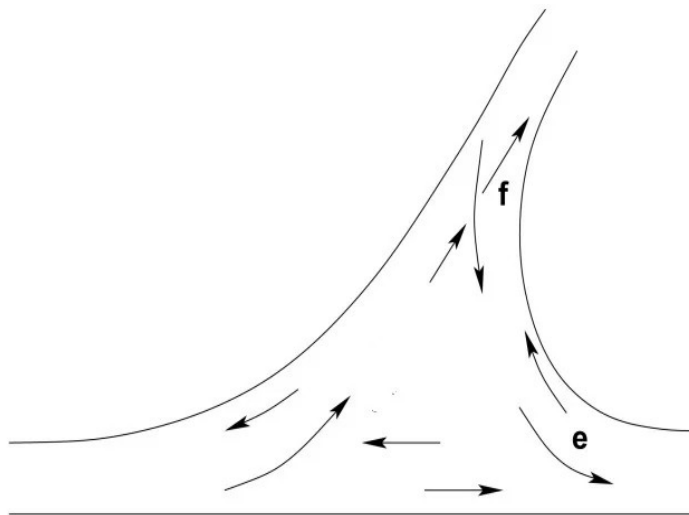
8. Differentiate between channelized and un-channelized intersections. State basic requirement of an intersection and mention how road marking and traffic signs enhance safety at a channelized intersection. [2+2+1=5]

Answer:

Road intersection is defined as an area where two or more roads join or cross within which are included with the roadway and roadside facilities for traffic movement in that area. It is desirable that the intersection roads meet at or nearly at right angles.



Channelization for Y Intersections



Un-Channelization for Y Intersections

The differences between channelized and un-channelized intersections are as follows:

Channelized intersection	Un-channelized intersection
i. It is the intersection which directs the traffic to definite paths by the means of traffic markings and island.	i. It is the intersection without island for directing traffic into definite path.
ii. It is achieved by introducing islands into the intersectional area which reduces the total conflict area available in the un-channelized intersection.	ii. In this type of intersection there is no provision of any direction island or central island.
iii. It helps to channelize the turning traffic, control their speed and angle of approach and decrease conflict area at the intersection.	iii. Traffic from all the approaching roads intermingles and proceed to their desired path.
iv. This intersection has better control on the traffic entering and leaving the intersection.	iv. They are most dangerous and inefficient, cheaper and has lowest grade.

The basic requirements of an intersection:

- i. There should be adequate visibility for all the approaching vehicles on all converging roads.
- ii. Width of pavement at radius for turning should be sufficiently provided.
- iii. Area of conflict should be reduced to minimum by introducing properly designed traffic island and channelizing island.
- iv. Sudden change of path should be provided.

Road markings and traffic signs enhances safety at a channelized intersection as:

- i. Channelization helps vehicles to move to definite path.
- ii. Angle of merging and diverging maneuvers can be kept minimum.
- iii. Points of conflict can be separated.
- iv. Speed control devices can be installed to force the vehicles to reduce their speed before entering the intersection.

9. For a hill road in geologically fragile zone in Nepal, discuss geometric design consideration (design speed, sight distance, gradient, hairpin bends). How do geological conditions affect alignment selection and slope protection? [5+5=10]

Answer:

Hilly region of Nepal covers the mountainous & steep terrain having a cross slope of 25 to 60% and more. The road passing through the hilly terrain with a cross slope of 25% or more is generally termed as Hill Road. Road alignment in Nepal's hilly terrain is governed by a combination of geological, topographic, drainage, economic, and geometric design factors. The steep, unstable slopes and complex geology of the Himalayas and Middle Hills make the design and construction of hill roads more critical.



Geometric design considerations in hill roads:

i. Design speed:

- Design speed is selected according to terrain and road class.
- In steep mountainous terrain, lower design speeds are adopted to improve safety.
- Typical values adopted are:
 - Mountainous terrain: 20–40 km/h
 - Rolling terrain: 40–60 km/h
- Lower speeds allow smaller curve radii and reduce excavation.

ii. Sight distance:

Adequate sight distance is essential because visibility is often restricted by sharp curves and steep terrain.

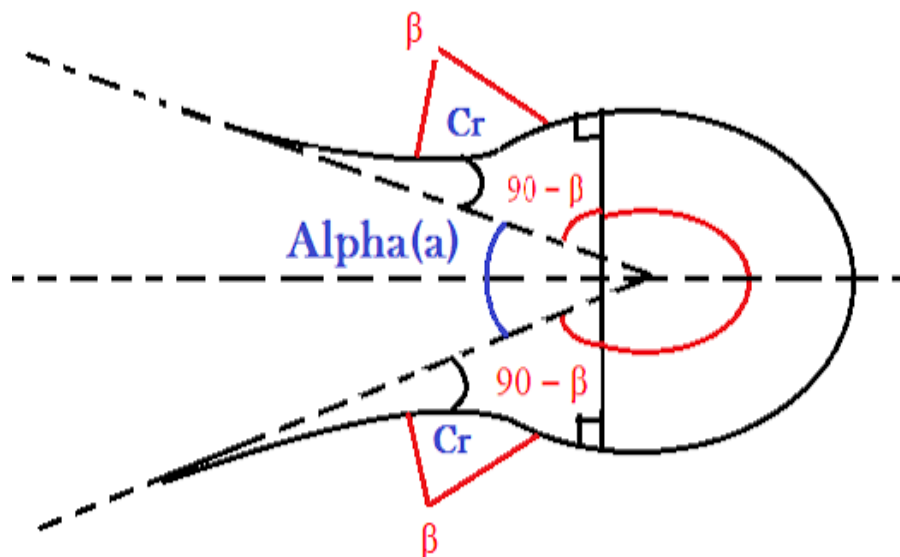
- Stopping Sight Distance (SSD): Minimum requirement throughout the road.
- Overtaking Sight Distance (OSD): Provided only where terrain permits.
- At blind curves, convex mirrors, warning signs, and speed restrictions should be provided.

iii. Gradient:

- Gradients should be kept as gentle as practicable to facilitate safe vehicle operation.
- Desirable gradient: 5–7%
- Maximum ruling gradient: 7–10% (depending on standards and terrain).
- Exceptional gradient may be permitted for short lengths only.
- Frequent grade compensation is provided on sharp curves.

iv. Hairpin bends:

- Hairpin bends are those provided in hill road alignment introduced when it becomes necessary to attain height without covering substantial horizontal distance.
- In hairpin bends,
 - a. Minimum design speed= 20 kmph
 - b. Minimum radius of curvature = 15 m
 - c. Minimum length of transition curve = 15 m
 - d. Maximum longitudinal gradient= 4%
 - e. Maximum superelevation =10%



Effect of geological conditions on alignment selection and slope protection:

i. Weak and weathered rock:

- Road alignment is avoided through weak and weathered rock masses to reduce slope failures and excessive maintenance.
- Suitable slope protection includes turfing, bio-engineering, geotextiles and gentle slope flattening.

ii. Faults and fractured rocks:

- These zones cross at the shortest possible length and is avoided to pass through faults and shear zones to minimize stability.
- Suitable slope protection measures include rock bolting, shotcrete, wire mesh and rockfall barriers.

iii. Landslide-prone areas:

- Road alignment is shifted away from the active and old landslide zones to improve long-term stability.
- Suitable slope protection measures include retaining structures, soil nailing and vegetation.

iv. Soil type and bearing capacity:

- Routes with good quality of rock and soil is preferred to ensure stable road subgrade.

- Reinforced earth walls and breast walls can be constructed for slope protection.

v. Groundwater and seepage:

- The areas with excessive groundwater, springs are avoided to reduce the slope stability.
- It should be maintained at least 1.2 m below the road subgrade.
- Catch-water drains, sub-surface drains, horizontal drains, side drains are provided for slope protection.

10. Explain the concept of slope protection using bio-engineering techniques. Describe different bio-engineering methods used for slope protection in hilly roads. [5+5=10]

Answer:

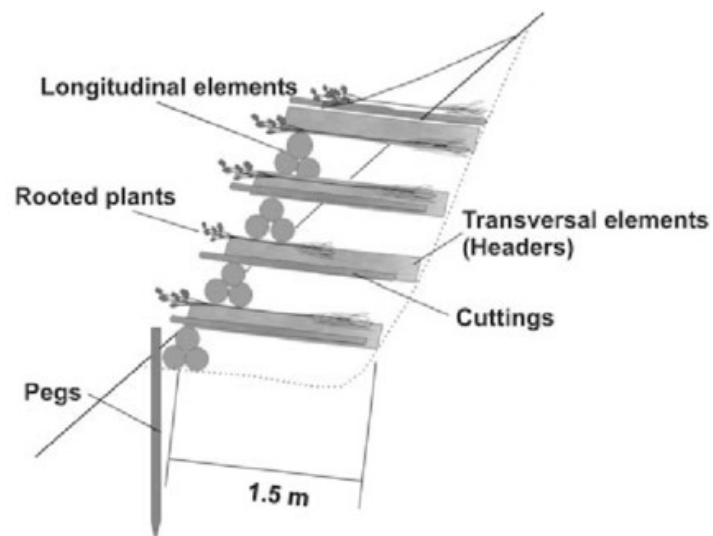
Stabilization of slope using vegetative systems with small civil engineering structures is known as bio-engineering. It combines biological measures (plants) with civil engineering works to improve slope stability in an environmentally friendly and cost-effective manner.

Principle of bio-engineering:

- The main concept of bio-engineering is the interaction between vegetative systems and small civil engineering structure.
- Plant roots bind soil particles and increase shear strength.
- Vegetation reduces the impact of rainfall and surface runoff.
- Plants absorb soil moisture, lowering pore water pressure.
- Natural cover protects slopes from erosion and weathering.

Example:

- Toe wall with bamboo plantation:



- Jute net with grass seedling:



Application of Bio-engineering helps to stabilize slopes, control erosion, and reduce landslide risks. It is widely used on hilly roads in Nepal because it is economical, environment-friendly and sustainable. The different bio-engineering methods used for slope protection in hilly roads are described below:

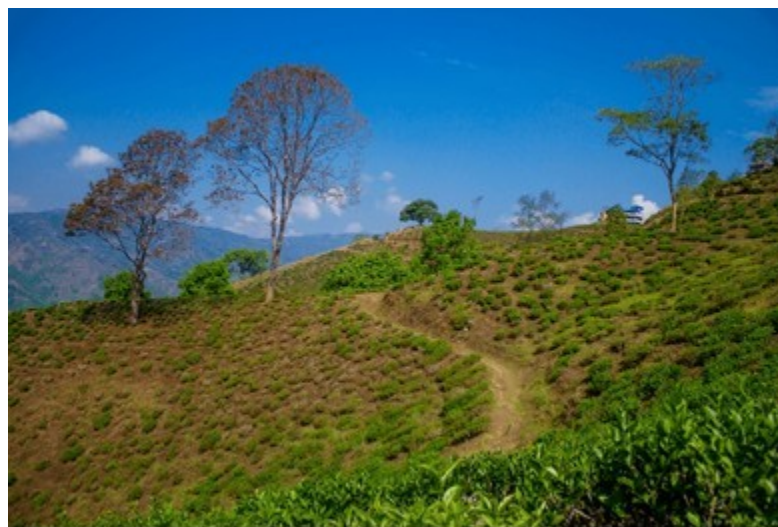
i. Turfing:

- Fast-growing grasses (e.g., Vetiver, Napier, Broom grass) are planted on exposed slopes.
- Dense root systems bind soil particles and reduce surface erosion caused by rainfall.
- Suitable for newly cut slopes and embankments.



ii. Shrubs and tree plantation:

- Deep-rooted shrubs and trees are planted to reinforce the soil.
- Roots increase soil shear strength and improve long-term slope stability.
- Also reduce runoff and improve the surrounding ecosystem.



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iii. Fascines:

- Bundles of live branches are tied together and placed along contour lines.
- They slow runoff, trap sediments, and encourage vegetation growth.
- Commonly used on moderately steep slopes.



iv. Check dams:

- Small barriers made from bamboo, brushwood, or live stakes are built across gullies.
- Reduce runoff velocity, trap sediments, and prevent gully erosion.
- Particularly useful in drainage channels.



v. Palisades:

- Wooden stakes with live cuttings are driven into unstable slopes.
- Provide immediate support while developing a living root network.
- Suitable for shallow failures and eroded slopes.



vi. Jute netting:

- Natural fiber mats made of coconut (coir) or jute are spread over exposed soil.
- Protect the soil from rain splash and erosion until vegetation becomes established.
- Eventually decompose naturally.



vii. Gabion walls and crib walls:

- Gabion or timber crib walls are filled with soil and planted with grasses or shrubs.
- Combine structural support with biological stabilization.
- Suitable for high cut slopes and retaining structures.

Section-D (25 Marks)

11. Describe the various methods of population forecasting techniques. Discuss the factors that influence the per capita water demand in an urban area of Nepal. [5+5=10]

Answer:

It is very essential to forecast the design year population so that a water supply project can be designed to meet the needs of community not only for the present but up to the end of design period. Otherwise, there might be inadequacy of water and the project may fail to fulfill its objective.

The various method of population forecasting techniques are given below:

i. Arithmetic Increase Method:

- This method is based on the assumptions that the population is increasing at constant rate.
- It is suitable for old, stable cities with slow growth.
- Advantages: Simple and easy to apply.
- Limitations: Not suitable for rapidly growing towns.

$$P_n = P_0 + \Im I$$

Where :

P_n = Forecast population

P_0 = Present population

I = Average population increase per decade

n = Number of decades

ii. Geometric Increase Method:

- This method is suitable for new and developing cities.

- Geometric Increase Method is based on the assumption that the percentage increase in population from decade to decade remains constant.
- Advantages: Reflects exponential growth.
- Limitations: May overestimate future population.

$$P_n = P_0(1+r)^n$$

Where :

r = Average growth rate per decade

iii. Incremental Increase Method:

- This method is an improvement over two methods of forecasting: arithmetical and geometrical.
- Considers both the average population increase and the increment in that increase.
- More realistic than the arithmetic method.
- Advantages: Suitable for towns with gradually changing growth.

$$P_n = P_0 + nC + \frac{n(n+1)}{2} i$$

Where :

i = Average incremental increase

C = average increase

iv. Decreasing Rate of growth Method:

- If the complete growth of a very old city is plotted, it will be seen that the curve has S-shaped.
- In this method, average decrease in percentage increase is worked out and is then subtracted from the latest percentage increase for each successive decade.
- Suitable for mature cities approaching saturation.
- Advantages: Produces more realistic estimates for developed cities.

There are various factors that influence per capita water demand of water. They are:

- Climate conditions:
 - The amount of water required in hot and dry place due to use of air coolers, air conditioners, sprinkling water in lawns, gardens, more washing and cleaning etc.
- Size and type of community:
 - Bigger the size of community, higher the demand of water for the personal hygiene and recreational purpose.
- Living Standard of People:
 - Higher the living standard of people, higher will be the demand of water.
- Quality of water:
 - If the quality of water is good, then the demand of water will be higher and vice-versa.
- Pressure in supply:

- Higher the pressure, loss of water is more and higher will the demand of water, thus increasing the per capita demand of water.
- vi. System in supply:
 - There are two systems of supply either continuous or intermittent.
 - Water demand is less in intermittent supply of water due to many problems such as limited water, storage requirements etc.
- vii. Cost of water:
 - If the cost of water is higher, demand of water is also higher.

12. What type of environmental study are conducted in Nepal? How do you identify the types of environmental study for the energy sector projects? [3+7=10]

Answer:

As per EPA 2076 and EPR 2077, there are three types of environmental studies required before any construction activities. They are:

i. Basic Environmental Study (BES):

- BES is conducted to identify the possible environmental impacts of a small-scale project and to recommend measures for preventing those impacts.
- Need of BES in road and bridge projects:
 - a. BES recommends simple mitigation measures to reduce the adverse effects.
 - b. It is done for projects that are likely to have only minor environmental effects.
 - c. It ensures compliance with environmental regulations at minimum cost and time.
- According to EPA2076 and EPR2077, energy projects requiring BES are:
 - i. **Transmission Lines:** Construction of electricity transmission lines up to 66 KV.
 - ii. **Hydropower:** Projects with a capacity less than **1 MW**.

ii. Initial Environment examination (IEE):

- IEE is conducted to determine whether a proposed medium scale project or activity will cause adverse environmental impacts during its implementation, and to identify measures to avoid, reduce, or mitigate such impacts.
- If IEE Report submitted is approved by concerned authority, the project is implemented.
- Need of IEE in road and bridge projects:
 - a. IEE determines whether the impacts can be mitigated through practical measures.
 - b. it is required for medium scale projects that has moderate effect on environment.
 - c. It also helps in assisting decision-makers in selecting environmentally acceptable alternatives.
 - d. It evaluates the impacts on land, water resources, forests, wildlife and socio-economic conditions.
- According to EPA2076 and EPR2077, energy projects requiring an IEE are:
 - i. **Hydropower:** Projects with a capacity between **1 MW and 50 MW**.
 - ii. **Transmission Lines:** Construction of electricity transmission lines of **132 kV or higher** capacity, or new outdoor substations tapped from existing **220 kV or higher** lines.
 - iii. **Thermal:** Generation from mineral oil or gas between **1 MW and 5 MW**.
 - iv. **Renewable Energy:**
 - a. Solar and wind energy projects between **1 MW and 10 MW**.

- b. Bio-energy projects between **0.5 MW and 2 MW**.
- c. Biogas plants with a capacity exceeding **1,500 cubic meters**.
- d. Cogeneration (e.g., from sugar industry bagasse)

iii. Environmental Impact assessment (EIA):

- It is a detailed study or evaluation to be conducted to ascertain the potential adverse environmental impacts of a proposed project or activity and to identify measures to avoid, mitigate, or compensate for such impacts.
- Need of EIA in road and bridge projects:
 - a. EIA provides detailed assessment of physical, biological, social, cultural and economic impacts.
 - b. EIA is required for large-scale development projects with significant environmental impacts that cannot be adequately assessed through a Brief Environmental Study (BES) or Initial Environmental Examination (IEE).
 - c. EIA also examines project alternatives, including design and selection of route alignment.
 - d. It also develops comprehensive environmental management and monitoring plans. Prior to the implementation of a proposal, it is necessary the proponent shall prepare an environmental management plan.
- According to EPA2076 and EPR2077, energy projects requiring an EIA are:
 - i. **Hydropower:** Projects with a capacity exceeding **50 MW**.
 - ii. **Thermal/Nuclear:** Any coal or nuclear-based electricity generation project with a capacity exceeding **1 MW**, or mineral oil/gas projects exceeding **5 MW**.
 - iii. **Renewable Energy:**
 - a. Solar and wind energy projects exceeding **10 MW**.
 - b. Bio-energy projects exceeding **2 MW**.